

Section Overview

What You Will Learn

Lists

Searching in Lists

Exception Handling

Slices

Ranges

For Loop

While Loop

Lists

Lists

- A list is a data type that holds an ordered collection of items.
- The items can be of various data types.
- You can even have lists of lists!

Creating Lists

```
list_name = [item_1, item_2, item_N]
```

```
list_name = []
```

```
list_name[index]
```

```
animals = ['man', 'bear', 'pig']  
print(animals[0])  
print(animals[1])  
print(animals[2])
```

man
bear
pig

```
animals = ['man', 'bear', 'pig']  
print(animals[0])  
animals[0] = 'cat'  
print(animals[0])
```

man

cat

```
animals = ['man', 'bear', 'pig']  
print(animals[-1])  
print(animals[-2])  
print(animals[-3])
```

pig
bear
man


```
animals = ['man', 'bear', 'pig']  
animals.append('cow')  
print(animals[-1])
```

COW

```
animals = ['man', 'bear', 'pig']  
animals.extend(['cow', 'duck'])  
print(animals)
```

```
more_animals = ['horse', 'dog']  
animals.extend(more_animals)  
print(animals)
```

```
['man', 'bear', 'pig', 'cow', 'duck']
```

```
['man', 'bear', 'pig', 'cow', 'duck', 'horse', 'dog']
```

```
animals = ['man', 'bear', 'pig']  
animals.insert(0, 'horse')  
print(animals)
```

```
animals.insert(2, 'duck')  
print(animals)
```

```
['horse', 'man', 'bear', 'pig']  
['horse', 'man', 'duck', 'bear', 'pig']
```

Slices

Slices

```
list[index1:index2]
```

```
list[:index2]
```

```
list[index1:]
```

```
animals = ['man', 'bear', 'pig', 'cow', 'duck', 'horse']

some_animals = animals[1:4]
print('Some animals: {}'.format(some_animals))

first_two = animals[0:2]
print('First two animals: {}'.format(first_two))

first_two_again = animals[:2]
print('First two animals: {}'.format(first_two_again))
```

```
Some animals:      ['bear', 'pig', 'cow']  
First two animals: ['man', 'bear']  
First two animals: ['man', 'bear']
```

```
animals = ['man', 'bear', 'pig', 'cow', 'duck', 'horse']  
  
last_two = animals[4:6]  
print('Last two animals: {}'.format(last_two))  
  
last_two_again = animals[-2:]  
print('Last two animals: {}'.format(last_two_again))
```


Last two animals: ['duck', 'horse']

Last two animals: ['duck', 'horse']

String Slices

```
part_of_a_horse = 'horse'[1:3]  
print(part_of_a_horse)
```

or

Finding an item in a list.

```
animals = ['man', 'bear', 'pig']  
bear_index = animals.index('bear')  
print(bear_index)
```

1

Exceptions

```
animals = ['man', 'bear', 'pig']  
cat_index = animals.index('cat')  
print(cat_index)
```

```
Traceback (most recent call last):  
  File "exception_example.py", line 2, in <module>  
    cat_index = animals.index('cat')  
ValueError: 'cat' is not in list
```

Exception Handling

Exception Handling

```
animals = ['man', 'bear', 'pig']  
try:  
    cat_index = animals.index('cat')  
except:  
    cat_index = 'No cats found.'  
print(cat_index)
```

No cats found.

Loops

Looping through a list

```
for item_variable in list_name:  
    # Code block
```

```
item_variable = list[0]  
item_variable = list[1]  
item_variable = list[N]
```



```
animals = ['man', 'bear', 'pig']  
for animal in animals:  
    print(animal.upper())
```

MAN

BEAR

PIG

While Loop

```
while condition:  
    # Code block
```

```
animals = ['man', 'bear', 'pig', 'cow', 'duck', 'horse']  
  
index = 0  
  
while index < len(animals):  
    print(animals[index])  
    index += 1
```

```
man  
bear  
pig  
cow  
duck  
horse
```

Sorting and Ranges

```
animals = ['man', 'bear', 'pig']
sorted_animals = sorted(animals)
print('Animals list:          {}'.format(animals))
print('Sorted animals list:   {}'.format(sorted_animals))
animals.sort()
print('Animals after sort method: {}'.format(animals))
```

```
Animals list:          ['man', 'bear', 'pig']
Sorted animals list:   ['bear', 'man', 'pig']
Animals after sort method: ['bear', 'man', 'pig']
```

```
animals = ['man', 'bear', 'pig']  
more_animals = ['cow', 'duck', 'horse']  
all_animals = animals + more_animals  
print(all_animals)
```

```
['man', 'bear', 'pig', 'cow', 'duck', 'horse']
```

```
animals = ['man', 'bear', 'pig']  
print(len(animals))  
animals.append('cow')  
print(len(animals))
```

3

4

Ranges

```
for number in range(3):  
    print(number)
```

0

1

2


```
for number in range(1, 3):  
    print(number)
```

1

2

```
for number in range(1, 10, 2):  
    print(number)
```

1
3
5
7
9

```
animals = ['man', 'bear', 'pig', 'cow',  
           'duck', 'horse', 'dog']  
for number in range(0, len(animals), 2):  
    print(animals[number])
```

man
pig
duck
dog

Section Summary

Summary

Lists are created using comma separated values between square brackets. The format is:

```
list_name = [item_1, item_2, item_N]
```

Summary

Items in a list can be accessed by index. List indices are zero based. The format is:

```
list_name[index]
```

Access items from the end of the list by using negative indices. The last item in a list is:

```
list_name[-1]
```

Summary

Add items to a list by using the `append()` or `extend()` list methods.

Access a portion of a list using a slice. The

format is: `list_name[start, stop]`

The list `index()` method accepts a value as a parameter and returns the index of the first value in the list or an exception if the value is not in the list.

Summary

Loop through a list using a for loop. The format is `for item_variable in list_name:` followed by a code block.

The code block in a while loop executes as long as the condition evaluates to true. The format is `while condition:` followed by a code block.

Summary

To sort a list, use the `sort()` list method or the built-in `sorted()` function.

The built-in `range()` function generates a list of numbers. The format is: `range(start, stop, step)`

Unhandled exceptions cause Python programs to terminate. Handle exceptions using `try/except` blocks.